

西南财经大学本科期末考试试题册（B）

（2025—2026学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVASE）(英) 命题教师：王磊

适用对象（年级专业）：2025 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[√]
- 2、本套试题共 二 道大题，共 5 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 资料1本[√] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证并放置于桌面左上角，以备查验。
 2. 领取试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名等信息填写完整。
 4. 所有答案均需填写在答题纸上，填写在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. Which of the following is **not** a valid Java identifier?

- A. `_userName` B. `123test`
C. `$price` D. `studentName123`

2. What is the output of the following code?

```
int x = 5, y = 3;  
System.out.println(x > y ? "x is bigger" : "y is bigger");
```

- A. x is bigger B. y is bigger
C. true D. false

3. What is the value of sum after the following loop?

```
int sum = 0;  
for (int i = 1; i <= 10; i++) {  
    if (i % 3 == 0)  
        continue;  
    sum += i;  
}
```

- A. 37 B. 45
C. 55 D. 0

4. Analyze the following code, which description is correct?

```
class Test {  
    public static int add(int x, int y) {  
        return x + y;  
    }  
}
```

```

    }

    public static double add(double x, double y) {
        return x + y;
    }

    public static void main(String[] args) {
        System.out.println( add(3, 5) + " " + add(2.5, 3.5) );
    }
}

```

- A. The program outputs "8 6.0"
 - B. The program outputs "8 6"
 - C. The program outputs 14.0
 - D The program outputs 14
5. Which of the following arrays is initialized **correctly**?
- A. `int[] arr = new int[5]{1,2,3,4,5};`
 - B. `int[] arr = {1,2,3,4,5};`
 - C. `int arr[] = new int[];`
 - D. `int arr[] = {1,2,3,4,5,...};`
6. Which of the following is **correct** about static methods in a class?
- A. They can access instance variables directly
 - B. They can be called without creating an object of the class
 - C. They can be overridden by subclass methods
 - D. They must be called using an object reference
7. What is the output of the following code?

```

class Animal {
    public void sound() {
        System.out.println("Animal sound");
    }
}

```

```

    }
}
class Dog extends Animal {
    public void sound( ) {
        System.out.println("WoWo");
    }
}
public class Test {
    public static void main(String[] args) {
        Animal a = new Dog();
        a.sound();
    }
}

```

- A. Animal sound
- B. Runtime error
- C. nothing
- D. WoWo

8. Which of the following is **true** about abstract classes?

- A. They can be instantiated directly
- B. They can contain only abstract methods
- C. A class can inherit multiple abstract classes
- D. They can contain concrete methods and abstract methods

9. Analyze the following program, which description is **correct**?

```

class ExceptionTest {
    public static void main(String[] args) {
        try {
            int[] nums = {1, 2, 3};
            System.out.println(nums[3]);
            System.out.print("Success ");
        }
    }
}

```

```
        catch (ArrayIndexOutOfBoundsException e) {  
            System.out.print("Array index error ");  
        }  
    }  
}
```

- A. The program outputs "Success"
- B. The program outputs "Array index error"
- C. The program outputs " Array index error Success "
- D. The program has a compile error

10. Which description about Java FX layout is **correct**?

- A. VBox arranges components in a horizontal row
- B. FlowPane wraps components to the next line when there's not enough space
- C. GridPane will not change the original size of the component
- D. BorderPane's center region cannot be empty

Part II: Write Programs.(total 70 pts)

1. (10pts) Write a program that output all the factors of a given non-negative integer. (e.g. for integer 18, its factors include 1,2,3,6,9,18)
2. (15pts) A supermarket is running a promotion. If the shopping amount exceeds 99 yuan, all items receive a 10% discount; if it exceeds 199 yuan, all items receive a 15% discount; and if it exceeds 299 yuan, all items receive a 20% discount. Please write a Java program for the supermarket to calculate the discounted consumption amount.
3. (20pts) Define an abstract class Vehicle with:
 - Member variables: brand[品牌], price[价格].
 - A constructor to initialize the variables.
 - An abstract method start() that returns a void data type.

Define two subclasses Car and Motorcycle that extend Vehicle:

- Each has a member variable speed [速度]
- Each has a constructor to initialize variables, in which the speed is initialized with 0
- Each overrides the start() method: Car print "Car started", Motorcycle print "Motorcycle started", when invoke the method.
- Each has a method speedup(int e), to increase the object's speed (with e) as well as to print current speed.

Write a test program to create objects of both subclasses, and call their start() method to start the vehicle, the speedup() method to change their speed. The main() method is given as below.

```
public static void main(String[] args) {  
    Car v1= new Car("Jeep", 32.5);  
    Motorcycle v2= new Motorcycle("Toyoto", 10.2);  
    v1.start();  
    v1.speedup(100);  
    v2.start();  
    v2.speedup(60);  
}  
}
```

4. (25 pts) Please design a Java GUI program, which can find all the prime integers within any specified range. The GUI program may run like below. When the user inputs the lower bound (integer type) and the upper bound (integer type), and click the “Display Prime” button, it will display all the prime integers in the third TextFiled object.

Requirements:

- ◆ It should have suitable Java FX GUI components and layout.
- ◆ It should contain at least one event handling process.

